

COMICCRAFT – PROJECT PLANNING PHASE

AI Comic Story Creator using Gemini Models

1. Planning Overview

The Project Planning Phase organizes the development of ComicCraft into milestones and activities. The project documentation divides the work into model selection and architecture, core functionality development, route development, frontend development, and deployment/testing.

2. Project Goal

The goal is to develop an end-to-end AI comic creation application that accepts a user's story preferences, generates a structured comic story and illustrations, presents the result through a web interface, and provides a downloadable PDF.

3. Development Milestones

Milestone	Main Activities	Planned Outcome
1. Model Selection and Architecture	1.1 Research and select generative AI model 1.2 Define application architecture 1.3 Set up development environment	Selected AI workflow, system architecture, and ready dev
2. Core Functionalities Development	2.1 Develop core functionalities 2.2 Implement FastAPI backend for routing and user input processing	AI generation modules and backend workflow
3. routes.py Development	3.1 Write main application logic in routes.py	Integrated application routes and workflow handling
4. Frontend Development	4.1 Design and develop user interface 4.2 Create dynamic Jinja2 templates	User input, comic preview, and export pages
5. Deployment	5.1 Prepare application for local deployment 5.2 Test and verify local deployment	Working local application and verified workflow

4. Milestone 1 – Model Selection and Architecture

Activity 1.1: Research and select appropriate generative AI models. The documentation identifies Gemini Flash for structured comic outlines, Gemini Pro for detailed narration and dialogue, and Stable Diffusion for comic-style illustrations.

Activity 1.2: Define the architecture around the Frontend, FastAPI Backend, and AI Integration components.

Activity 1.3: Set up Python, pip, a virtual environment, required libraries, project structure, and the local Uvicorn server.

5. Milestone 2 – Core Functionalities Development

Activity 2.1: Implement the main AI and processing functions: generate_outline(), generate_story(), generate_image(), build_comic_layout(), and save_pdf().

Activity 2.2: Implement the FastAPI backend to receive user input and coordinate the complete comic-generation workflow.

6. Milestone 3 – routes.py Development

The planning sequence includes development of the main application routes: homepage, comic generation, JSON comic generation, image testing, and export-success handling. The routes connect user requests with the corresponding AI and export functions.

7. Milestone 4 – Frontend Development

Activity 4.1: Design the user interface for story prompt, character name, setting, story tone, and art style. Develop the comic preview and export-success views.

Activity 4.2: Use FastAPI's Jinja2 templating to dynamically bind generated panel data, images, descriptions, captions, and narration to the frontend.

8. Milestone 5 – Deployment and Testing

Activity 5.1: Prepare local deployment using a virtual environment, install requirements, configure environment variables, and start the FastAPI application.

Activity 5.2: Start the Uvicorn server, access the application locally, and verify the available API endpoints through the FastAPI documentation.

9. Task Dependencies

The planned development follows a dependency-based sequence: model and architecture selection → environment setup → core AI functions → FastAPI routes → frontend templates → integration → local deployment and testing.

10. Resource Planning

The documented technical resources include Python and pip, FastAPI, Uvicorn, Jinja2, Google Generative AI, Hugging Face Diffusers and Stable Diffusion, PyTorch-related dependencies, FPDF, Pillow, Accelerate, and Git. API credentials are planned to be managed through environment variables or a .env file.

11. Testing and Verification Plan

Testing is planned after local deployment. The application homepage and FastAPI interactive documentation are accessed locally. The documented endpoints include /generate-comic/json, /generate, /export-success, and /test-image.

12. Project Deliverables

The planned deliverables are a working ComicCraft web application, AI-generated five-panel comic content, generated panel illustrations, a browser-based comic preview, and an exportable PDF comic.

13. Risk and Mitigation Planning

The project documentation indicates the need for exception handling around API failures and JSON processing. A modular architecture separates story generation, image generation, layout, and export so individual stages can be tested and maintained separately.

14. Planning Summary

The Project Planning Phase provides a structured development path from initial AI model selection through architecture, core implementation, routing, frontend integration, deployment, and verification. This sequence supports the end-to-end ComicCraft workflow described in the project documentation.

Reference: ComicCraft - AI Comic Story Creator using Gemini Models (provided project documentation).